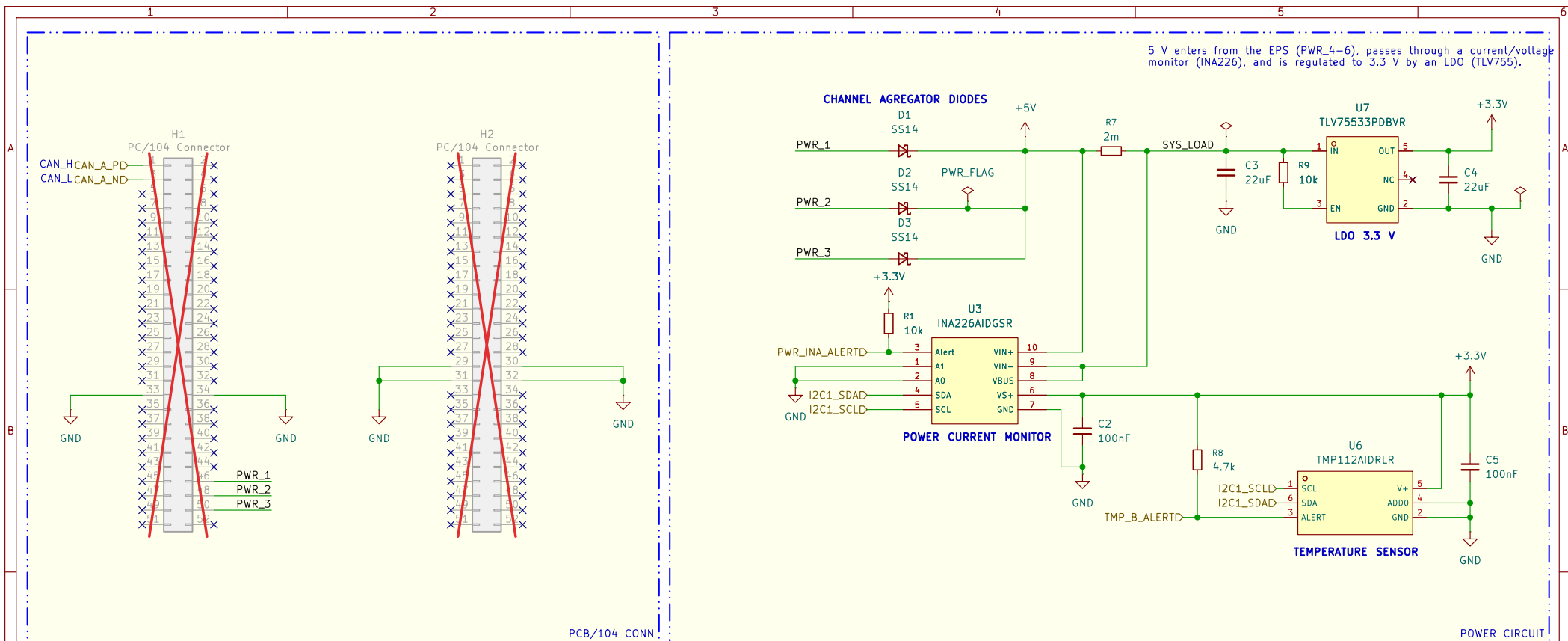
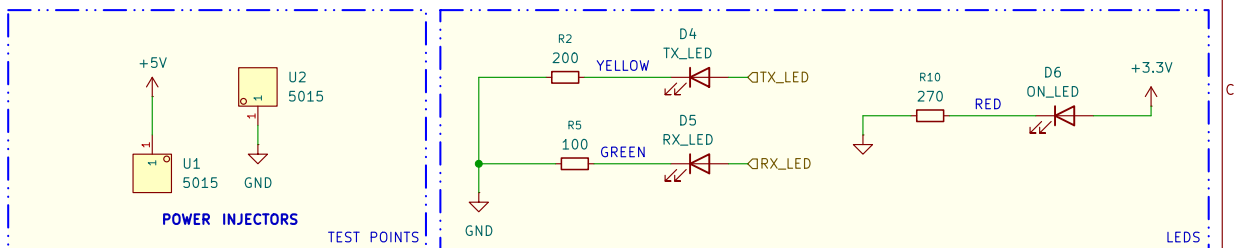




POWER SUPPLY

- Main power conditioning stage of the module.
- Receives regulated 5 V input from the EPS through the PC/104 connector and distributes power to the onboard subsystems.
- Schottky diode OR-ing network used to combine multiple EPS power channels (PWR_1, PWR_2, PWR_3).
- Prevents reverse current flow between channels and increases power redundancy.
- High-side current and voltage monitoring circuit based on the INA226 sensor.
- Measures input current consumption and bus voltage from the EPS power line through a low-ohmic shunt resistor.
- Low-dropout voltage regulator generating the main 3.3 V rail used by the digital and RF subsystems.
- Converts the incoming 5 V supply into a regulated low-noise output voltage.
- Monitors the temperature of the power regulation section and surrounding circuitry.
- Connected through the primary I2C bus and capable of generating thermal alert interrupts.
- Multiple PC/104 connector pins used for redundant power injection into the module from the spacecraft EPS.
- Visual indicators used for power presence and communication activity monitoring.
- Includes power-on, TX, and RX status indication.



nt	Name/Signature		Date		Title		
	Drawn Rodrigo Andrade		09/06/2026		TET Module		
	Checked						
	Approved (1)				Inatel CubeSat Design Team		
	Approved (2)				Date 2026-06-09	Revision v1r1	Size A4
					Software KICad E.D.A. 10.0.2	Sheet 2/3	Code TET-v1r1

